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Full Name	Nikolas Johannes Jahn
Institution Name	Ziehenschule
Country	Germany
Category	Junior
Class/Year	Grade 12
Registered Email Address	nikolasj.jahn@gmail.com
Submission Date	September 21, 2026

A Reinforcement-Learning Approach to Joint Ambulance and Defibrillator-Drone Dispatch for Rural Emergency Medical Response in Germany

Research Objectives

This research is about getting a defibrillator to a patient faster by using ambulances and delivery drones together instead of separately. Reinforcement learning, a method where software learns the best strategy through repeated trial and error rather than being given fixed rules in advance, is the tool this study uses to make that joint decision automatically.

Objectives: (1) Create a combined ambulance/drone dispatch simulator and train a reinforcement-learning policy on it using realistic response-time data; (2) Implement a dispatch policy that corresponds to the real-world approach and compare the two policies in terms of their performance in achieving time-to-defibrillation and meeting Hilfsfrist, Germany's legal maximum ambulance response time, on 10,000 simulated emergency events. (3) Explore how the performance of each policy varies depending on fleet size and location of the drone base.

Introduction

A clinically urgent clock: Every minute of delay in administering defibrillation following a cardiac arrest means a drop in survival rates of about 10 to 12 percent [6]. This is why speed is regarded as so critical for emergency-response services. It is a question of life or death, not just service quality.

A concrete setting: A district such as Vogelsbergkreis, sparsely populated and rural, in Hesse a little over an hour from Frankfurt, is exactly the scenario where distance works against that clock: long stretches between villages and the nearest ambulance station mean minutes can be lost before a vehicle even leaves its base.

Where drones and reinforcement learning enter separately, but not together: Delivery drones have already been proposed and prototyped as a way to get medical equipment to a patient faster than a ground vehicle can [9], and unmanned aerial vehicles are now a well-established tool in emergency and disaster response more broadly [7], [8], [10]. Separately, reinforcement learning has proven capable of solving complex sequential decision-making problems, from Atari games [2] to a wide range of control and resource-allocation tasks [11]. What hasn't been shown is a single system that uses reinforcement learning to jointly decide, in real time, whether to send an ambulance or a drone as multiple emergencies happen at once.

Why this is personal: The researcher's own years spent building and troubleshooting drones and custom PCBs, including the kind of motor and ESC failures that make hardware unreliable in the field, are exactly why this proposal treats a dispatch policy that has only ever been tested in a clean simulation with caution rather than assumed success.

The proposed research addresses that gap directly.

Literature Review

Clinical stakes: Response time is a hard outcome (rather than a soft service metric) because international resuscitation guidelines state that the probability of survival following cardiac arrest decreases by 10 to 12 percent with each minute delay in defibrillation [6].

Reinforcement learning as a proven method: Reinforcement learning has been shown to reach strong performance on complex sequential decision-making tasks [2], and has since been surveyed and applied across a wide range of control and resource-allocation problems [11].

Proximal Policy Optimization, the specific algorithm used in this proposal, is among the most widely adopted and reliable of these methods [1].

Drones as an emergency-response tool: Unmanned aerial vehicles are an established and actively growing tool for disaster and emergency response [7], [8], [10], including specifically for delivering medical equipment faster than ground vehicles can reach a patient [9].

Gaps in Current Literature: Reinforcement learning and drone-based emergency response have each matured as separate research threads, but no reviewed source combines them: even the broad surveys of drone-based emergency response [7], [8], [10] and of reinforcement learning applications [11] cited above make no mention of a single learned policy that jointly and adaptively dispatches both ambulances and medical delivery drones. That combination, a new pairing of two separately mature tools rather than an incremental extension of either, is the specific gap this proposal addresses.

Research Question

Given a set of stochastic cardiac-arrest calls, can a single reinforcement-learning-based dispatch policy assign both ambulances and drones carrying defibrillators to respond to them (regarded as a pool of resources) such that mean and worst-case scenario (95th-percentile) time-to-defibrillation is reduced compared to a fixed, distance-based dispatch policy reflective of current practice? Finally, how well does this new policy perform in terms of the German legal ambulance response-time limit, the Hilfsfrist?

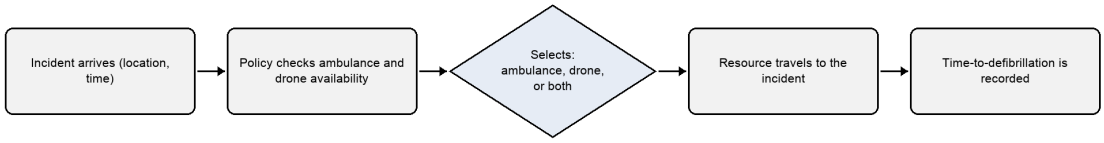
The main parameters of the experiments are as follows: Independent variable: The dispatch policy of the ambulance and drone (learned vs fixed-threshold). Dependent variable: Time-to-defibrillation with respect to the fixed-threshold baseline and the Hilfsfrist standard

Methodology

This is a computational simulation study using reinforcement learning; no physical hardware, human subjects, or real patient data are involved.

Figure 1

The dispatch policy's decision loop, evaluated once per incident



1. Simulation environment: extend an existing, standard reinforcement-learning environment interface [5] to add a second dispatchable resource type, defibrillator-carrying drones, alongside ambulances. The road network modeled is Vogelsbergkreis, a rural district in Hesse, built from OpenStreetMap data via OSMnx, a tool that turns real-world map data into a format a computer program is able to navigate [3], [12].

2. Incident generation: relies on synthetic cardiac-arrest events produced from arrival-rate and response-time distributions drawn from established resuscitation and EMS literature [6], and no real patient records are used. The evaluation uses the same 10,000 simulated incidents for both policies, with incidents generated from multiple randomized incident-arrival seeds to assess consistency.

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3. Algorithm: Proximal Policy Optimization (PPO), a widely used algorithm that learns a good decision-making strategy through repeated trial and error [1], implemented in Python using Stable-Baselines3, a ready-made software library of reinforcement-learning algorithms.

4. Baseline: a fixed, distance-threshold-based dispatch rule. The baseline is the simplest rule that reflects how dispatch decisions are made manually today using distance-based thresholds. It is implemented independently to compare the performance of the learned policy to the baseline.

5. Variables: Independent: Dispatch policy (reinforcement-learning vs. fixed-threshold). This is the main comparison. Fleet size and drone-base placement. Detailed description of this variable is provided below as a sensitivity analysis. Dependent: Time-to-defibrillation and Hilfsfrist compliance rate. Controlled variables: Incident arrival distribution and road network, keeping the fleet size and base locations constant for the main comparison.

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6. Analysis: statistical comparison of the two policies' resulting response-time distributions across the simulated incidents, using a Mann-Whitney U test, a standard statistical method for checking whether two sets of results are meaningfully different rather than just different by chance [4], on median response time, and comparison of each policy's compliance rate against Germany's Hilfsfrist response-time standard. The trained policy is then re-evaluated under three additional fleet-size and drone-base-placement configurations to measure how sensitive its performance is to those choices, addressing Objective 3.

7. Ethics and safety: No human subjects or real patient data are used, only synthetic events with incident and response times designed to be consistent with published statistics.

8. Risks and mitigation: Simulation-fidelity risk: Avoid arbitrary values for distribution of incidents and response times by using clinically plausible distributions calibrated in relation to the available clinical and EMS literature [6]. Reinforcement-learning training instability risk: Ensure that the environment works correctly in the simple fixed-threshold baseline case before expanding it to the full joint-dispatch setting. Scope risk due to increased complexity: Avoid a fully decentralized multi-agent model. Instead, use this as a single joint-policy approach.

Timeline

Month 1 (Weeks 1-4): Build the simulation. Weeks 1-2: Extend the reinforcement-learning environment interface and build the Vogelsbergkreis road network using OSMnx (Methodology #1). Weeks 3-4: Build and calibrate the synthetic cardiac-arrest incident generator to the literature on established resuscitation and EMS (#2).

Month 2 (Weeks 5-8): Define the experiment and build the baseline. Week 5: Conclude the independent/dependent/controlled variable definitions and experimental design (#5). Week 6: Document the ethics and safety protocol confirming no real patient data or human subjects are involved (#7). Weeks 7-8: Implement the fixed-threshold dispatch rule representative of current practice as the comparison baseline (#4).

Month 3 (Weeks 9-13): Train and validate the policy. Weeks 9-11: Implement and train the PPO policy (#3). Weeks 12-13: Validate the trained policy against the fixed-threshold baseline on a small-scale run to ensure the policy has not been overly sensitive to the training set before scaling up (#8).

Month 4 (Weeks 14-17): Full evaluation and write-up. Weeks 14-15: Run the full 10,000-incident evaluation for both policies. Week 16: Statistical comparison, Hilfsfrist compliance

analysis, and the fleet-size/drone-base-placement sensitivity analysis (#6). Week 17: Finalize results and write-up.

Expected Outcomes

If successful, this study would achieve: A trained joint-dispatch policy and its fixed-threshold-baseline comparison tested on the same 10,000 simulated incidents in the Vogelsbergkreis road network. An answer to whether the learned policy can beat today's one-shot predict-and-threshold approach on average and worst-case scenario time-to-defibrillation. A measure of how well the two policies conform to Germany's Hilfsfrist standard, as well as a sensitivity analysis on fleet size and placement of drone bases. The result either way will be a good, functional solution for rural EMS planners, which can be directly translated into more people surviving a cardiac arrest.

Conclusion

What remains an open question is whether a single policy for reinforcement learning can make a difference in responding to cardiac-arrest events in rural areas. The proposal is to treat ambulances and defibrillator-carrying drones as a single fleet for joint dispatch. This would address all three goals stated above (a working joint-dispatch simulation, a fair comparison against a fixed-threshold rule representing current practice, and a clear understanding of how performance varies with fleet size and drone location). If the reinforcement-learning approach does better than the fixed-rule approach current systems use, this would be the first demonstration that thinking about dispatch as a sequential decision problem works better than that fixed-rule approach. As the evaluation is a simulation, the obvious next step would be to evaluate it in the field before considering any real deployments. And finally, once a deployment has been found, the idea can be extended to other types of intervention (such as naloxone or epinephrine) and to other parts of Germany outside of Hesse. The ultimate aim is simple: provide life-saving equipment to a patient faster, wherever they happen to collapse.

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